One trip can have several correct descriptions

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Displacement / ES: desplazamiento / PT: deslocamento

Δ x = x f − x i . Its sign depends on the axis you chose; the physical motion does not.

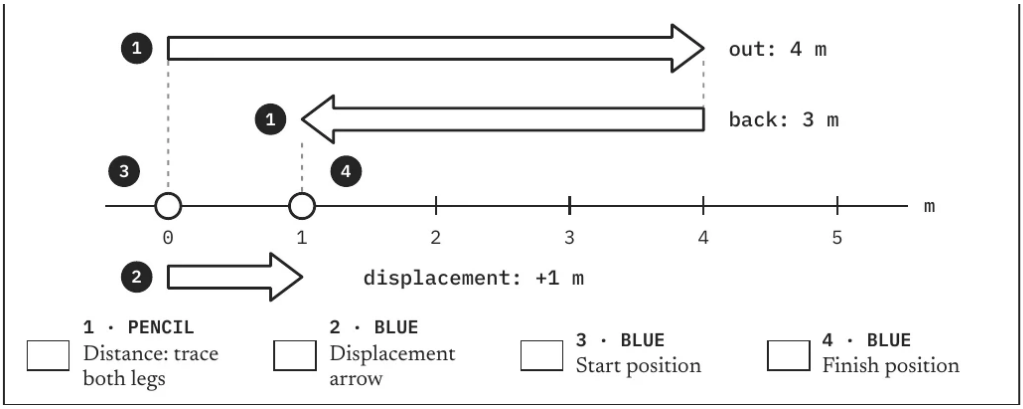
Also on your note page: Position, Distance, Scalar / vector.

B READ THE RELATIONSHIP

Displacement = final position − initial position

As you read, find: start · finish · direction.

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

Start at −3 m, move to +2 m, then finish at −1 m. Find the distance and the displacement, with units.

Given x i = −3 m turn at +2 m x f = −1 m

Rule distance = Σ |leg| Δ x = x f − x i

Work |2 − (−3)| + |−1 − 2| = 5 + 3 −1 − (−3)

Answer distance 8 m displacement +2 m

E SAVE FOR THE FINAL MEETING

XJ's position only means something in a reference frame: an origin, axes, and a sign convention. Its displacement, not its path length, decides where it ends up.

Read the physics: Displacement

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Draw an origin: this is zero. Choose which way counts as positive. Now a position tells both people the same location.

02 HOW IT WORKS

Trace every part of the trip to find distance. For displacement, go straight from start to finish. Subtract the starting position from the final position. Keep the sign.

03 WHAT TO CHECK

The object can travel a long distance and end close to its start. If it returns to its start, displacement is zero.

04 TRY THE EXAMPLE FROM YOUR NOTES

Start at −3 m, move to +2 m, then finish at −1 m. Find the distance and the displacement, with units. distance 8 m displacement +2 m.

Model check: Displacement = final position − initial position.

05 CONNECT TO XJ

XJ's position only means something in a reference frame: an origin, axes, and a sign convention. Its displacement, not its path length, decides where it ends up.

SAY IT BEFORE YOU WRITE

Which two positions belong in the subtraction? The path is ___. The change from start to finish is ___.

Build the word. Draw the idea.

STUDENT CHOICE / FRONT / OPENING MEETING Name _____

A FRAYER MODEL

Displacement / ES: desplazamiento / PT: deslocamento

Say it in any language. Draw or use a few words. Use the notes to check your idea.

Meaning	How can you tell?
To me, displacement means …	The cart turns around. Which two positions do you need to find its displacement? I compare where it ___ with where it ___.
Example or sketch	**Non-example**
I see it when …	This looks close, but …

B DRAW WITH YOUR TEACHER

Put start, turn, and finish on a number line. Label distance on the path and displacement on the shortcut.

Labels you can use: start · finish · direction.

Choose zero, direction, and scale.

Tell a partner: The path is ___. The change from start to finish is ___.

On assignment: Displacement

SECTION E / BACK / FINAL MEETING ONLY Name _____

A survey rover tracks XJ's model from +2 to +8 to +5 km. Dispatch needs both mileage and final location.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Route engineer**

Draw the two legs for the mileage log and a single arrow for the return crew. Give dispatch the two different numbers.

☐ **Radio dispatcher**

Guide a new driver to the finish. Give the trip length, then its change in position, and tell the driver which number locates the finish from the start.

Word bank: start · finish · direction.

Try: "Our desk recommends ___. This mark shows ___."

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

"We first said ___. Now we ___ because ___."

Editor's check: Explain what changes if your partner reverses the positive axis.